

SVKM's NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING
SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2021-22

Programme: B. Tech (Data Science)

Year: II

Semester: IV

Subject: Database Management System

Date: 08 June 2022

Marks: 100

Time: 10.00 am to 1.00 pm

Durations: 3 (Hrs)

No. of Pages: 3

Re-Examination (2021-22)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1		Answer briefly:	[20]
CO-4 ; SO-1 ; BL-2	a.	Write a short note on conflict and view serializability.	[5]
CO-1 ; SO-1 ; BL-2	b.	Explain different types of attributes with examples. Explain specialization and generalization with suitable examples.	[5]
CO-3 ; SO-1 ; BL-2	c.	Distinguish between primary and super keys. Explain Referential Integrity constraint with suitable examples.	[5]
CO-4 ; SO-1 ; BL-2	d.	Differentiate static hashing and dynamic hashing.	[5]

Q2 CO-1; SO-1; BL-1	a.	Define Data models and its types in detail.	[10]
CO-1; SO-2; BL-2	b.	Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted.	[10]
Q3 CO-3; SO-1; BL-3	a.	Consider the R(A B C D E F G H I J K L M) with Functional dependencies: { A → B C D E, E → F G H, I → J, A I → K, A L → M } i. Find the candidate key ii. Indicate all BCNF violations for R. iii. Decompose the relations into collections of relations that are in BCNF.	[15]
CO-3; SO-1; BL-3	b.	Explain the Fourth normal form and Fifth normal form with suitable examples.	[5]
Q4 CO-1; SO-1; BL-1	a.	Explain Three Tier architecture with suitable diagrams.	[10]
CO-2; SO-1; BL-3	b.	Explain the types of joins with examples.	[10]
Q5 CO-2; SO-2; BL-3	a.	Consider the following relational schema to write the queries in SQL. employee (employee name, street, city) works (employee name, company name, salary) company (company name, city) manages (employee name, manager name) i. List the CREATE TABLE statements with the attribute names, and appropriate data types. Also make sure to have the primary keys and foreign keys clearly defined. ii. Find the names and cities of residence of all employees who work for "First Bank Corporation".	[20]

		iii. Find the names, street addresses, and cities of residence of all employees who work for "First Bank Corporation" and earn more than \$10,000. iv. Find all employees in the database who do not work for "First Bank Corporation". v. Find all employees in the database who earn more than each employee of "Small Bank Corporation". vi. select T.company-name from company T where (select R.city from company R where R.company-name = T.company-name) contains (select S.city from company S where S.company-name = 'Small Bank Corporation') vii. Find the company that has the most employees. viii. Find those companies whose employees earn a higher salary, on average, than the average salary at "First Bank Corporation". ix. Modify the database so that "Jones" now lives in "Newtown". x. Give all managers of "First Bank Corporation" a 10 percent raise unless the salary becomes greater than \$100,000; in such cases, give only a 3 percent raise.	
Q6 CO-4; SO-1; BL-2	a.	What is the 2-phase locking protocol and strict two phase locking protocols? How does it guarantee serializability?	[10]
CO-4; SO-1; BL-2	b.	Why do we need concurrency control? Explain it. Illustrate Concurrent execution of transactions with examples?	[10]